# Mebrahtu F. Weldeghebriel

Department of Geosciences, Princeton University, Princeton, NJ 08544, USA
mweldeg@princeton.edu | Google Scholar


## RESEARCH INTERESTS

Low-temperature and isotope geochemistry · paleoclimatology · ancient seawater chemistry · long-term carbon cycle · evaporites, brines, carbonates and dolostones · critical mineral resources in sedimentary basins · ice-core geochemistry · development of in situ analytical methods.

## EDUCATION

**Binghamton University, State University of New York**, Binghamton, NY — Aug 2016 – Jul 2022
PhD, Geological Sciences

Thesis: *Secular variability in major, minor, and trace element composition of Phanerozoic and Neoproterozoic seawater: evidence from fluid inclusions in marine halite.* Advisor: Dr. Tim Lowenstein, Distinguished Professor.

**Eritrea Institute of Technology (EIT)**, Asmara, Eritrea — Sep 2006 – Jul 2011
BSc, Geology

Thesis: *Sedimentology and lithostratigraphic characteristics of the evaporite succession in the Colluli area, Danakil Depression, East African Rift, Eritrea.* Advisor: Dr. Nageshwar Dubey, Professor.

## RESEARCH EXPERIENCE

**Princeton University** — Gordon and Betty Moore Postdoctoral Research Fellow — Sep 2026 – Present
- Expanding the reconstructions of ancient ocean chemistry and dating the Antarctic ice cores.

**Princeton University** — NSF Postdoctoral Research Fellow — Sep 2024 – Sep 2026
- Developing a new chronometer based on $^{87}Sr/^{86}Sr$ ratios to date stratigraphically disturbed ice more than 1 million years old in Antarctica's blue ice areas, using LA-ICP-MS analysis of trapped sea salts.
- Producing high-resolution elemental maps of impurities in folded Antarctic blue ice to identify layering and stratigraphic orientation.

**Princeton University** — Harry H. Hess Postdoctoral Research Fellow — Sep 2022 – Sep 2024
- Improved column chemistry and analytical methods for lithium isotope analysis of carbonates and fluid inclusions in halite using ICP-QQQ-MS.
- Investigated the behavior of Li concentration and isotopic composition during evaporation of modern seawater.
- Reconstructed the lithium isotope composition of Neoproterozoic and Phanerozoic seawater from fluid inclusions in marine halite.
- Extended the lithium isotope record of seawater to the past 3 billion years using marine dolostones.
- Conducted mass-balance modeling of the mechanisms controlling long-term change in seawater lithium and the evolution of the global carbon cycle.

**Binghamton University** — Research Assistant — Aug 2016 – Jul 2022
- Developed an analytical technique for major, minor, and trace element analysis of fluid inclusions in marine halite combining LA-ICP-MS and cryo-SEM-EDS.
- Documented the chemical composition of ancient seawater from more than 1,371 fluid inclusions in >131 halites from ~32 Phanerozoic and Neoproterozoic marine evaporite basins.
- Investigated the drivers of long-term secular oscillations in seawater chemistry.
- Studied the sedimentology and geochemistry of recent evaporite sequences in the Dead Sea as a modern analog to deep hypersaline basins in the geological record.

**Eritrea Institute of Technology** — Research Assistant — Sep 2011 – Jun 2015
- Conducted field, petrographic, and geochemical analysis of Neoproterozoic granitoids in Eritrea (2013–2014).
- Performed GPS geodetic surveys of the Danakil Block and Afar Rift in collaboration with MIT (2012–2014).
- Deployed broadband seismometers for lithospheric tomography and for monitoring the 2011 Nabro eruption.

## GRANTS AND FELLOWSHIPS

**NSF Office of Polar Programs Postdoctoral Research Fellowship (OPP 2420291).** *A novel method for high-resolution chemical imaging and dating of old ice cores using LA-ICP-MS.* **Principal Investigator** (proposal conceived and written by the PI). $359,487 (2024–2026).

**Gordon and Betty Moore Foundation Postdoctoral Fellowship.** *Reconstructing ancient ocean chemistry and dating the Antarctic ice cores.* Postdoctoral Fellow. $60,000 (2026–2027).

**Harry H. Hess Postdoctoral Research Fellowship**, Department of Geosciences, Princeton University. *Reconstructing lithium isotope composition of Phanerozoic and Proterozoic seawater; mass-balance modeling of long-term change in seawater chemistry.* Postdoctoral Fellow. $147, 838 (2022–2024).

**Graduate research grants:** Mineralogy, Geochemistry, Petrology and Volcanology (MGPV) Division, Geological Society of America, $2,500 (2019); Society of Economic Geologists Foundation, $4,700 (2018); American Association of Petroleum Geologists Foundation American Association of Petroleum Geologists Foundation, $3,000 (2018).

## PEER-REVIEWED PUBLICATIONS

**Weldeghebriel, M. F.**, Lowenstein, T. K., 2023. Seafloor hydrothermal systems control seawater [$Li^+$]: evidence from fluid inclusions. *Science Advances* 9, eadf1605.

**Weldeghebriel, M. F.**, Lowenstein, T. K., Xia, Z., Li, W., 2023. Plate tectonic control of strontium concentration in Phanerozoic and Neoproterozoic seawater: evidence from fluid inclusions in marine halite. *Geochimica et Cosmochimica Acta* 346, 165–179.

**Weldeghebriel, M. F.**, Lowenstein, T. K., García-Veigas, J., Cendón, D. I., 2022. [$Ca^{2+}$] and [$SO_4^{2-}$] in Phanerozoic and terminal Proterozoic seawater from fluid inclusions in halite: the significance of Ca-$SO_4$ crossover points. *Earth and Planetary Science Letters* 594, 117712.

**Weldeghebriel, M. F.**, Lowenstein, T. K., García-Veigas, J., Collins, D., Sendula, E., Bodnar, R. J., Graney, J. R., Cendón, D. I., Lensky, N. G., Mor, Z., Sirota, I., 2020. Combined LA-ICP-MS and cryo-SEM-EDS: an improved technique for quantitative analysis of major, minor, and trace elements in fluid inclusions in halite. *Chemical Geology* 551, 119762.

Yang, Y., Liu, Y., Pogge von Strandmann, P. A., Jin, Z., Galy, A., Ye, C., **Weldeghebriel, M. F.**, Yan, Z., He, J., Gou, L. F., Deng, L., 2026. Decoupling of Neogene seawater lithium isotopes from uplift-driven weathering. *Nature Communications* 17, 6524.

Xia, Z., Li, S., Hu, Z., Bialik, O., Chen, T., **Weldeghebriel, M. F.**, … Li, W., 2024. The evolution of Earth's surficial Mg cycle over the past 2 billion years. *Science Advances* 10, eadj5474.

Li, Y., Wang, Y. Y., Murphy, J. G., **Weldeghebriel, M. F.**, Xiao, Y., 2024. Evaluating evaporites as potential archives of lake and seawater lithium isotopic ratio: an experimental study using various natural saline fluids. *Chemical Geology* 122326.

Lowenstein, T. K., **Weldeghebriel, M. F.**, Sirota, I., Eyal, H., Mor, M., Lensky, N. L., 2021. Criteria for the recognition of clastic halite: the modern Dead Sea shoreline. *Sedimentology* 68, 2253–2269.

Ajrhough, S., Boutarouine, H., Lowenstein, T. K., **Weldeghebriel, M. F.**, Xia, Z., El Arabi, E. H., 2024. Evaporites and red beds of a syn-rift Atlantic series: evidence of continental depositional environment (Berrechid sub-basin, Morocco). *Journal of Sedimentary Research* 94, 750–767.

Sendula, E., Gill, B. C., Rimstidt, J. D., Lowenstein, T. K., **Weldeghebriel, M. F.**, García-Veigas, J., Bodnar, R. J., 2022. Redox conditions in Late Permian seawater based on trace metal ratios in fluid inclusions in halite from the Polish Zechstein Basin. *Chemical Geology* 596, 120794.

## MANUSCRIPTS IN REVIEW AND IN PREPARATION

**Weldeghebriel, M. F.**, et al. A 550-million-year record of the concentration and isotopic composition of lithium in seawater. *Science Advances*, in review.

Guillerm, E., et al. (incl. **Weldeghebriel, M. F.**) High-fidelity reconstructions of element concentrations in halite fluid inclusions with Brillouin–LA-ICP-MS. *Chemical Geology*, in review.

## SELECTED CONFERENCE PRESENTATIONS

**Weldeghebriel, M. F.**, Lowenstein, T. K., Murphy, J. G., Jurikova, H., Rae, J. W. B., Niespolo, E. M., Higgins, J. A. A 550-million-year record of the concentration and isotopic composition of lithium in seawater. Goldschmidt 2026, Montreal, Canada.

**Weldeghebriel, M. F.**, Fairuz, I., Higgins, J. A., Niespolo, E. M. High-resolution 2D impurity mapping of million-year-old Antarctic ice by cryo-LA-ICP-MS. Goldschmidt 2026, Montreal, Canada.

**Weldeghebriel, M. F.**, et al. Secular variations in Phanerozoic seawater lithium and links to Earth's climate states and the carbon–silicate cycle. Gordon Research Conference, 2025, Maine, USA. *(invited)*

**Weldeghebriel, M. F.**, Xia, Z., Li, W., Lonsdale, M., Smith, E. F., Blättler, C., Nadeau, M., Murphy, J. G., Niespolo, E. M., Higgins, J. A. A 3-billion-year lithium isotope record from marine dolostones. Goldschmidt 2025, Prague, Czech Republic. *(session convener)*

**Weldeghebriel, M. F.**, et al. Secular variation of lithium concentration and isotopic composition of Phanerozoic and Neoproterozoic seawater. Goldschmidt 2024, Chicago, USA. *(invited)*

**Weldeghebriel, M. F.**, Lowenstein, T. K., Higgins, J. A. Variability in strontium and lithium composition of ancient seawater from fluid inclusions in halite. EGU General Assembly 2024, Vienna, Austria. *(session convener)*

**Weldeghebriel, M. F.**, et al. Reconstructing the secular evolution of the lithium isotope composition of Phanerozoic and Neoproterozoic seawater from marine halite. GSA Annual Meeting 2023, Pittsburgh, USA.

**Weldeghebriel, M. F.**, Lowenstein, T. K., Higgins, J. A. Variability in strontium and lithium concentration of ancient seawater from fluid inclusions in halite. Chapman Conference on Hydrothermal Systems, 2023, Agros, Cyprus.

**Weldeghebriel, M. F.**, Lowenstein, T. K., Sirota, I., Eyal, H., Mor, Z., Lensky, N. Coupling halite deposition in the Dead Sea to environmental conditions. AGU Fall Meeting 2021, New Orleans, USA.

**Weldeghebriel, M. F.**, Lowenstein, T. K. Secular variability of Phanerozoic seawater strontium: evidence from fluid inclusions in marine halite. AGU Fall Meeting 2019, San Francisco, USA.

**Weldeghebriel, M. F.**, et al. Seafloor hydrothermal systems control seawater chemistry: evidence from fluid inclusions in halite. GSA Annual Meeting 2019, Phoenix, USA.

**Weldeghebriel, M. F.**, Lowenstein, T. K. Major, minor and trace element evolution of Phanerozoic seawater. GSA Northeastern Section 53rd Annual Meeting, 2018, Burlington, Vermont, USA.

**Weldeghebriel, M. F.** Sedimentation and lithostratigraphic succession of evaporites in the Danakil Depression, Colluli, Eritrea. 3rd World Young Earth Scientists Congress, 2014, Dar es Salaam, Tanzania.

## TEACHING EXPERIENCE

**Binghamton University** — Graduate Teaching Assistant Aug 2016 – May 2021

Taught laboratory sections of 15–30 students each semester for five years.

- GEOL 470/570 Geochemistry (2018, 2019, 2020, 2021)
- GEOL 111 Planet Earth (2018, 2019, 2020)
- GEOL 214 Earth's Interior (2018)
- GEOL 414/514 Climate and Paleoclimate (2017)
- GEOL 102 Geology of the Solar System (2016, 2017)

**Eritrea Institute of Technology** — Lecturer and Teaching Assistant Sep 2011 – Jun 2015

- GEOL 201 General Geology — **Lecturer** (2011, 2012, 2013, 2014); developed course materials, including lectures, laboratories, quizzes, and examinations.
- GEOL 232 Geological Map Interpretation — **Lecturer** (2012, 2013, 2014).
- GEOL 212 Introduction to Rock-Forming Minerals — **Lecturer** (2015)
- GEOL 461 Introduction to Geophysics — **Lecturer** (2014, 2015)
- GEOL 341/322/431 Field Mapping of Sedimentary, Igneous, and Metamorphic Rocks — TA (2013, 2014, 2015); led eight undergraduate field camps for more than 100 students.

## MENTORING AND RESEARCH SUPERVISION

- **Princeton University:** trained an undergraduate summer intern and two graduate students in cryo-cell LA-ICP-MS analysis of ice core, sample preparation for lithium isotope column chemistry and ICP-QQQ-MS analysis.

- **Binghamton University:** trained two graduate students, a postdoctoral researcher, and laboratory manager in LA-ICP-MS analysis of fluid inclusions in halite and gypsum.
- **Eritrea Institute of Technology:** supervised more than 40 undergraduates in field mapping, thin-section preparation, and senior thesis research. Alumni now work at the Geological Survey of Eritrea and in the exploration and mining industry or pursued graduate degrees in Europe and North America.

## PROFESSIONAL AND INDUSTRY EXPERIENCE

**Lithium Americas Corp.**, Nevada, USA — Exploration Intern Jun 2021 – Sep 2021

- Conducted field exploration of lithium-bearing sedimentary basins at 25+ sites across six U.S. states.
- Collected 500+ rock and sediment samples for geochemical analysis; mapped in the field and by desktop using ArcGIS Pro and QGIS.
- Synthesized field and laboratory data into technical reports; identified new lithium rich deposits.

**South Boulder Mines Ltd.**, Asmara, Eritrea — Exploration Intern Jun 2010 – Mar 2011

- Mapped evaporites at the Colluli Potash Project, Danakil Depression; logged potash drill cores and collected samples for geochemical analysis.

## HONORS AND AWARDS

- Graduate Student Excellence Award in Research, Binghamton University (2021).
- Graduate Student Research Award for excellence in research proposal, GSA MGPV Division (2019).
- Research Assistantship Scholarship Award (2021–2022); Teaching Assistantship Scholarship Award (2016–2021).
- Provost Doctoral Summer Fellowship, Binghamton University (2016–2020).
- Türkiye Scholarship, Dokuz Eylül University, Turkey (2015).
- Outstanding Student and Gold Medal Winner, College of Science, Eritrea Institute of Technology (2011).

## SERVICE, LEADERSHIP, AND OUTREACH

- **Coordinator**, Environmental Geology and Geochemistry Seminar (EGGS) Lecture Series, Department of Geosciences, Princeton University, 2025–present. Invite and host researchers whose work is of broad interest to the Geosciences, Atmospheric and Oceanic Sciences, and Ecology and Evolutionary Biology.
- **Session Convener**, "Seawater Chemistry Through Time: Reconstructions, Mechanisms, and Feedbacks," Goldschmidt 2026.
- **Session Convener**, "The Evolution of Seawater Chemistry Through Time," Goldschmidt 2025.
- **Session Convener**, "The Carbon Cycle as Reflected in Ocean Chemistry and Marine Records," EGU General Assembly 2024.
- **Peer reviewer** for *Nature Communications*, *Geology*, *Earth-Science Reviews*, *Journal of Geochemical Exploration*, and *Geological Quarterly*.
- **Community service**, Mathematics and Life Skills Teacher, Shambuko Junior and Secondary School, 2008–2009.

## SKILLS

**Analytical:** LA-ICP-MS (including cryo-cell configuration and LIBS), MC-ICP-MS, ICP-OES, electron microprobe, XRD, gravity-column chromatography, prepFAST, fluid-inclusion heating/freezing stage.

**Scientific:** fluid-inclusion and brine geochemistry, carbonate geochemistry, seawater chemistry, sedimentology and petrography, geochemical and mass-balance modeling, paleoclimate reconstruction, mineral-resource exploration.

**Software:** MATLAB, SILLS, Iolite, EQL-EVP, PHREEQC, FREZCHEM, Geochemist's Workbench, ArcGIS Pro, QGIS, Adobe Illustrator.

**Affiliations:** Geological Society of America, American Geophysical Union, Society of Economic Geologists, European Association of Geochemistry.

**Languages:** Tigrinya (native), English.